

Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

2024

Bachelor in Computer Applications

Course Title: **Digital Logic**

Code No: **CACS103** Semester: **I**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

2. Define Digital computer. Subtract: $([101010.101 - 1000100.001])$ using both 1's and 2's complement.
3. What do you mean by K-map? Explain the K-map with three variables. Simplify $(F(p,q,r,s) = \sum(3,4,7,8,14))$ with don't care conditions $(d(p,q,r,s) = \sum(1,6,9,13))$. Design the logic circuit using minimum NAND gates.
4. Differentiate between combinational and sequential logic circuits. Implement a full adder circuit using a decoder and two OR gates.
5. Define priority encoder. Explain 8-to-3 priority encoder in detail.
6. Explain the duality theorem with an example. Draw logic gates that implement the expression:
7. How does a flip-flop differ from a latch? Explain clocked SR flip-flop with logic diagram, truth table, characteristic table, and excitation table.
8. Write short notes on (any two): a) State reduction table b) Multiplexer c) Synchronous and Asynchronous counter

Group C

Attempt any TWO questions [2x10=20]

9. Differentiate between PAL and PLA. Design a combinational circuit with four inputs (BCD digit) and four outputs lines that generate the 2's Complements of the binary input pattern with circuit diagram, truth table, and block diagram.
10. Explain shift register with parallel load. Design a synchronous Mod-10 counter to count in the sequence 0,2,4,5,6,8 using T flip-flops.
11. Explain how race condition in JK flip-flop can be resolved. A sequential circuit with two D Flip-Flops (A, B), inputs (x, y) , and output (z) is defined by:
$$\begin{aligned} A(t+1) &= xy' + xB \\ B(t+1) &= x'B + xA \\ z &= A \end{aligned}$$